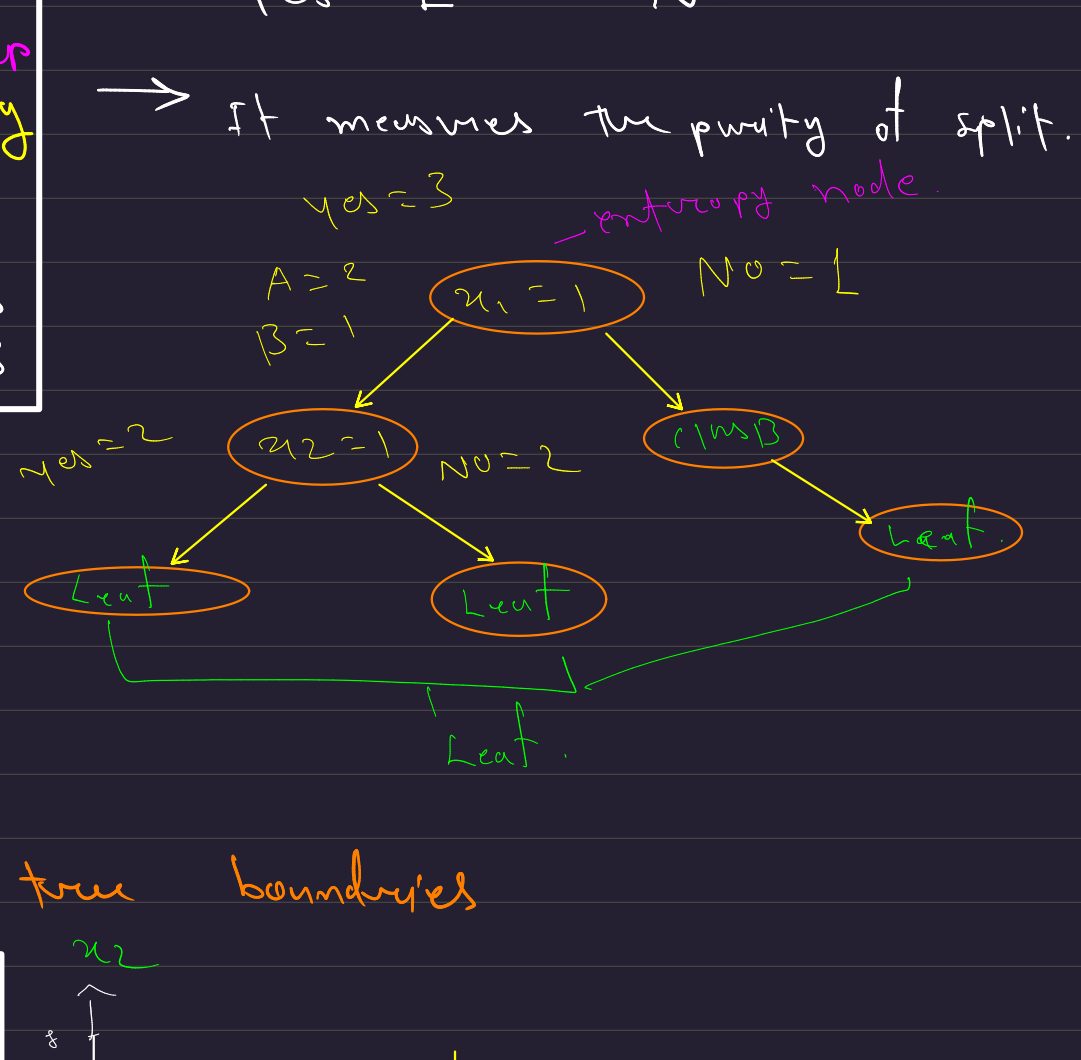


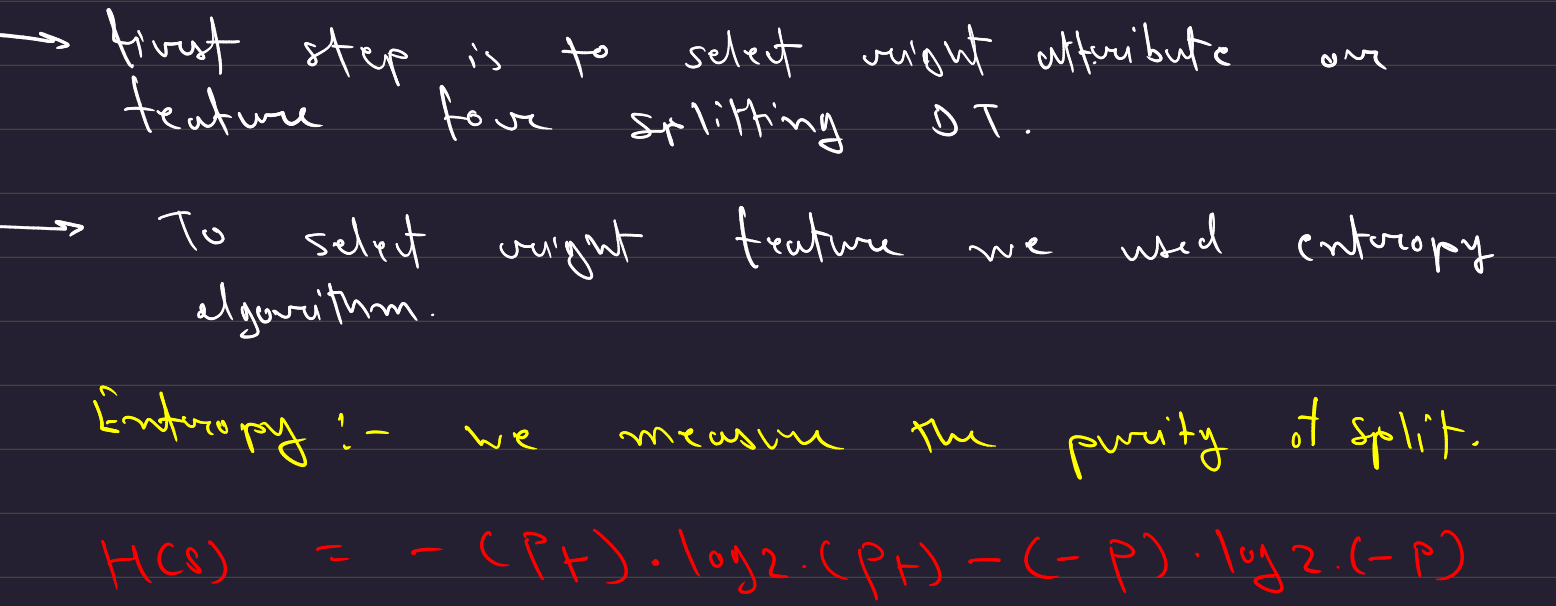
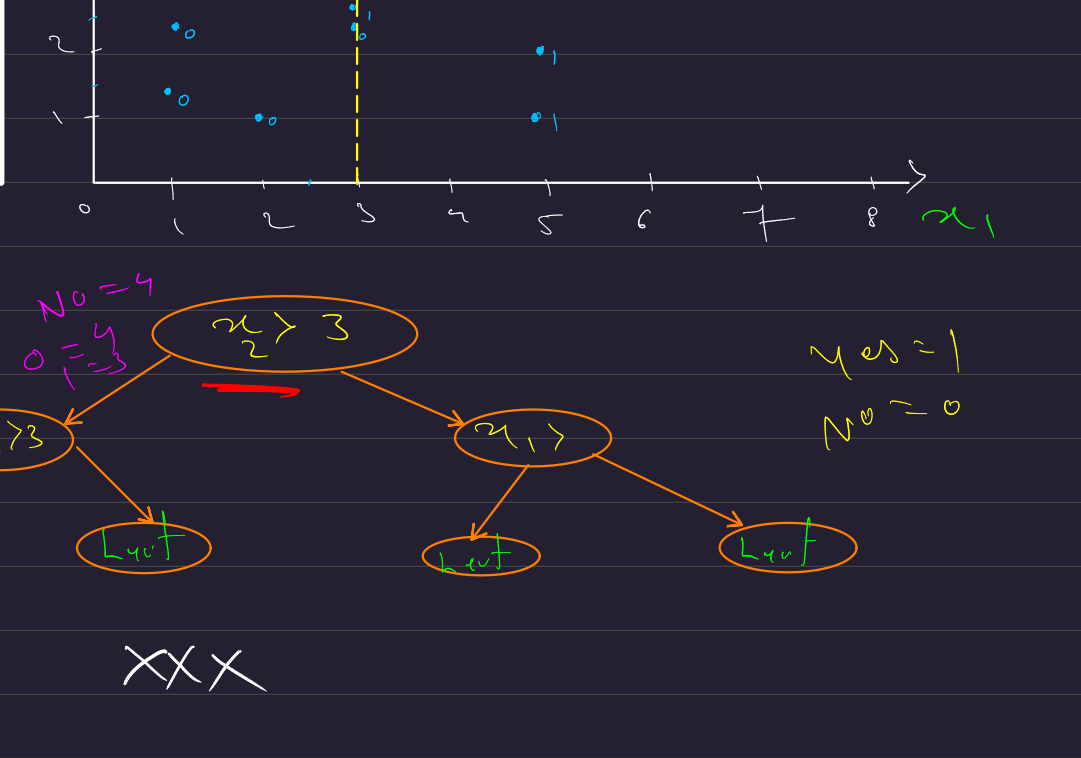
★ Decision Tree

Indep.			dep
x_1	x_2	x_3	y
1	1	1	A
1	1	0	A
0	0	1	B
1	0	0	B



★ Decision tree boundaries

x_2	x_1	Class
7	2.5	1
5.5	1.5	1
6	4.5	1
6	4	1
4.5	4.5	0
4.2	2.5	1
3.5	5	1
3.5	4.5	0
4	2.5	0
2.5	1	0
2.5	3	0
1.5	1	0
1	2	0
2.5	3	1
2	5	1
1	5	1

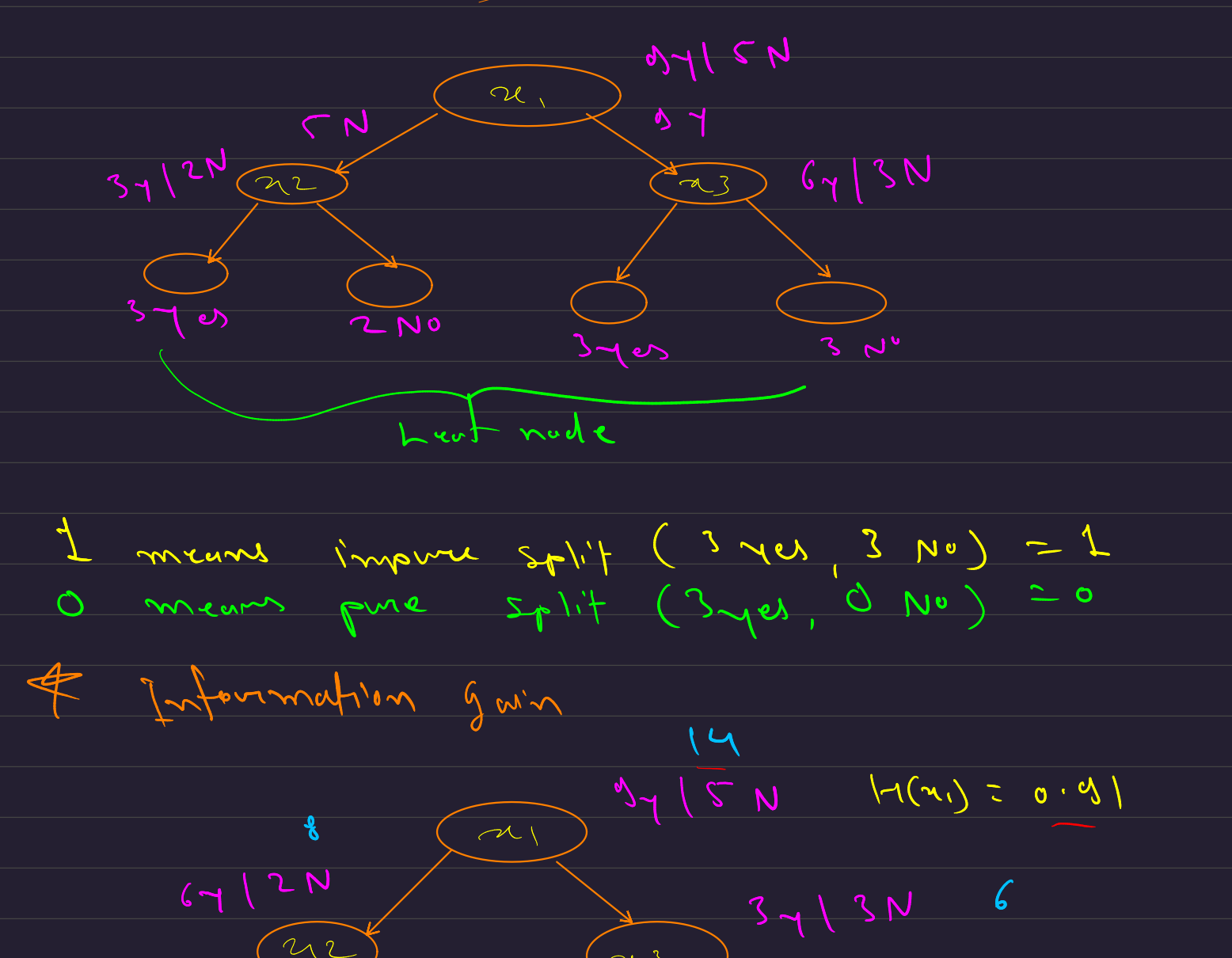


Entropy :- we measure the purity of split.

$$H(x) = -(P+) \cdot \log_2(P+) - (-P) \cdot \log_2(-P)$$

$$= \left(-\frac{3}{5}\right) \cdot \log_2\left(\frac{3}{5}\right) - \left(-\frac{2}{5}\right) \cdot \log_2\left(-\frac{2}{5}\right)$$

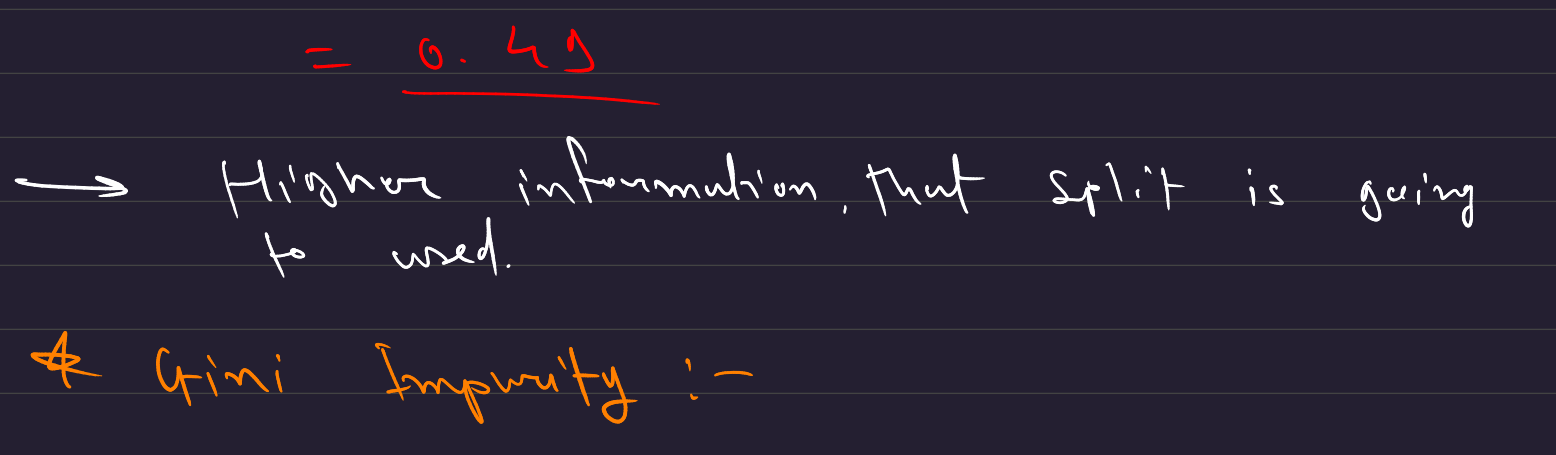
$$= \underline{0.78}$$



1 means impure split (3 yes, 3 No) = 1

0 means pure split (3 yes, 0 No) = 0

★ Information gain



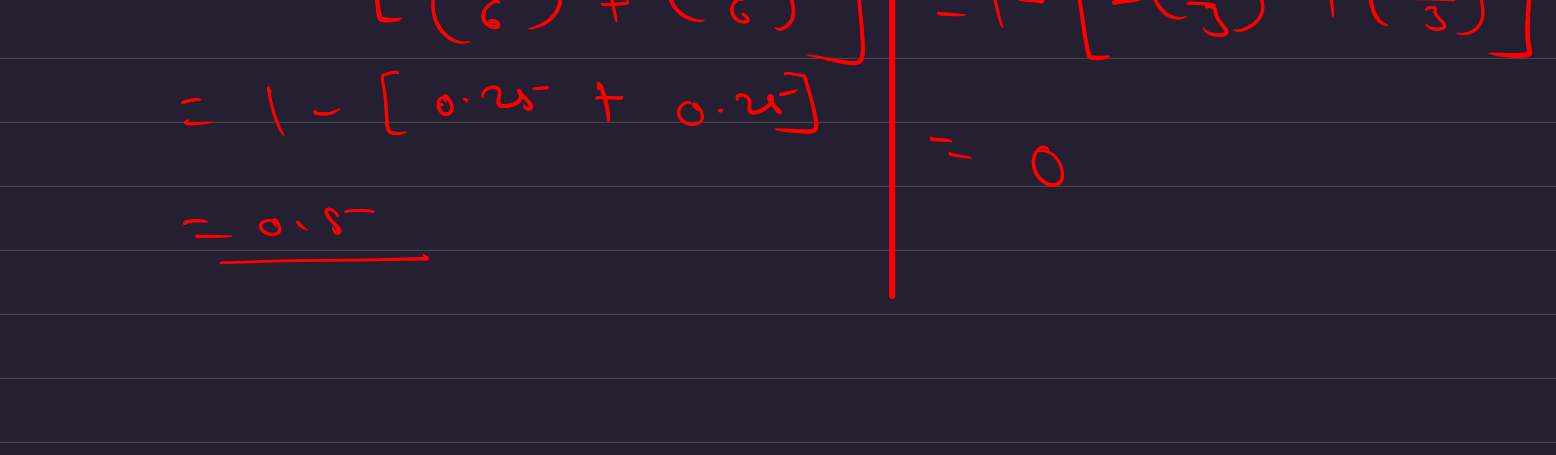
$$\text{gain} = H(x_1) - \frac{8}{14} \cdot H(x_2) - \frac{6}{14} \cdot H(x_3)$$

$$= 0.91 - \frac{8}{14} \cdot 0.81 - \frac{6}{14} \cdot 1$$

$$= \underline{0.49}$$

→ Higher information, that split is going to used.

★ Gini Impurity :-



$$GI = 1 - \sum_{i=1}^n (P_i)^2$$

$$= 1 - \left[-(P+)^2 + (P-)^2 \right]$$

$$= 1 - \left[-\left(\frac{3}{6}\right)^2 + \left(\frac{3}{6}\right)^2 \right]$$

$$= 1 - \left[0.25 + 0.25 \right]$$

$$= \underline{0.5}$$

$$= 1 - \left[-\left(\frac{3}{3}\right)^2 + \left(\frac{0}{3}\right)^2 \right]$$

$$= 0$$